

Koushik Varma Mudunuri

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WORK EXPERIENCE

Saayam For All – Full Stack Developer | Volunteer | Remote

Jul 2025 – Present

- Led a team of 3 volunteers to design and optimize secure APIs via API Gateway, achieving 100% integration coverage across 2 microservices.
- Strengthened authentication flows by implementing BCrypt hashing & validation layers, reducing security vulnerabilities by 30%.
- Counselled onboard new contributors by guiding local setups, troubleshooting errors, and resolving blockers — cutting setup/debugging time by 40% for the team.
- Took ownership of new features beyond initial scope, expanding from backend API work into UI/UX improvements, boosting volunteer request handling speed by 25%.

Tech Stack: Java, Spring Boot, AWS (Lambda, API Gateway), Docker, PostgreSQL.

Evolance – Software Engineer | Volunteer | Remote

Jun 2025 - Present

- Collaborated with the founder and two engineers to deliver backend APIs, improving request handling efficiency by 40% and supporting higher user load.
- Spearheaded RAG pipeline development and fine tuned LLMs with LangChain, boosting AI accuracy by 15% and reducing irrelevant responses.
- Facilitated code reviews and documentation updates, ensuring consistent coding standards and improving onboarding speed for new developers.
- Championed collaboration across product areas, boosting code test coverage by 20% and resolving 15 critical bugs in a single sprint.

Tech Stack: Python (FastAPI), React, PostgreSQL, JWT/OAuth 2.0, Docker, AWS ECS, Git

PROJECT EXPERIENCE

SkillMatch – Resume and Job Matching Platform (Ongoing)

- Designed a full stack job board platform (React, Node.js, PostgreSQL, AWS) with AI-driven resume–job matching and gRPC-based model integration.
- Integrated and fine tuned a PyTorch dual encoder model, achieving 91% top-3 accuracy and reducing false positives by 28%.

Tech Stack: PyTorch, Transformers, NLP, gRPC, PostgreSQL, Docker, AWS, React, Node.js

AI20Labs Document Q&A Assistant

- Implemented a Retrieval-Augmented Generation (RAG) pipeline using LangChain, LlamaIndex, and OpenAI, achieving 92% query accuracy with sub second response time.
- Incorporated mentor requested changes into the React chat UI to enhance usability and interface design.

Tech Stack: FastAPI, LlamaIndex, OpenAI LLMs, ChromaDB, React, LangChain, Docker, MongoDB, AWS.

Glossary Term Extractor Tool

- Created a glossary extraction pipeline using NLTK, lemmatization, and BabelNet APIs, reducing manual term creation by 2+ hours per document.
- Refined extraction accuracy through mentor guided feedback loops, improving precision by 15% and enabling LMS integration for scalable adoption.

Tech Stack: Python, NLTK, BabelNet API, Flask, Pandas

SKILLS

- Programming Languages & Paradigms:** Python, Java, JavaScript, TypeScript, C, OOP, Functional Programming
- Frameworks:** FastAPI, React.js, Next.js, Node.js, Express.js
- AI/ML:** PyTorch, Transformers, LLMs, CNNs, RNNs, LangChain, LlamaIndex, Hugging Face, TensorFlow
- Databases:** PostgreSQL, MySQL, MongoDB, ChromaDB
- Cloud/DevOps:** AWS (EC2, RDS, Lambda, ECS), Docker, Kubernetes, GitHub Actions
- Concepts:** RAG, LLM Fine Tuning, Resume Parsing, WebSockets, REST APIs, JWT/OAuth2, Microservices, CI/CD
- Tools:** Git, VS Code, Postman, Cursor, Linux/macOS

EDUCATION

University of Central Missouri, Warrensburg

MO, USA

M.S. in Computer Information Systems & IT, GPA: 3.9

Aug 2023 - May 2025

Coursework: Software Engineering, Enterprise Systems Design, Distributed Computing, Project Management, Advanced DB Management

Vishnu Institute of Technology, Bhimavaram

AP, India

B.S. in Computer Science Engineering, GPA: 3.5

Aug 2019 - May 2023

Coursework: Web Technologies, AI & ML, NLP, Data Structures & Algorithms, OOP (Python & Java), Operating Systems

CERTIFICATIONS & ACHIEVEMENTS

- AWS Cloud Internship:** Built and deployed cloud-native applications using EC2, Lambda, and RDS; gained practical experience in scalable distributed systems and DevOps pipelines.
- TCS CodeVita & CodeKaze:** Ranked Top 1% (1668/150K+) and Top 0.3% (453/140K+) globally among competitive programmers.
- Solved 400+ problems across LeetCode, Codeforces, and HackerRank, specializing in Data Structures, Algorithms, and Problem-Solving.